Unit 5

# Property and Loans

# Unit 5: Property and Loans

## Property

The Good, the Bad and the Ugly

Sydney property market

Property taxes

Property valuation

Human bias and property

Before you buy

Making an offer

After you buy

Property strategies

## Loans

Financial leverage

Loan mathematics

Loan features

Interest rates

Home loan tips

# Discussion

**Home prices are very high in Australia (and many other countries).**

**Average home price divided by average income is high.**

**Interest rates are high and banks are willing to lend less.**

**Rent is high. Food is expensive. The cost of living is high. It is hard to save.**

**It can be challenging for university graduates to get a good job.**

**What pressures do you feel from ‘society’ or family to buy a property?**

**How do you feel about the problem of ‘home affordability’?**

# General Advice Warning

**The content of this Unit is <u>not</u> personal financial advice for you**

**It covers general principles**

**… taught as part of university course based on the Australian system.**

**It does <u>not</u> take into account your situation, objectives or needs.**

**You are responsible to consider whether it is suitable**

**… for your personal circumstances.**

**You should consider obtaining financial and tax advice yourself.**

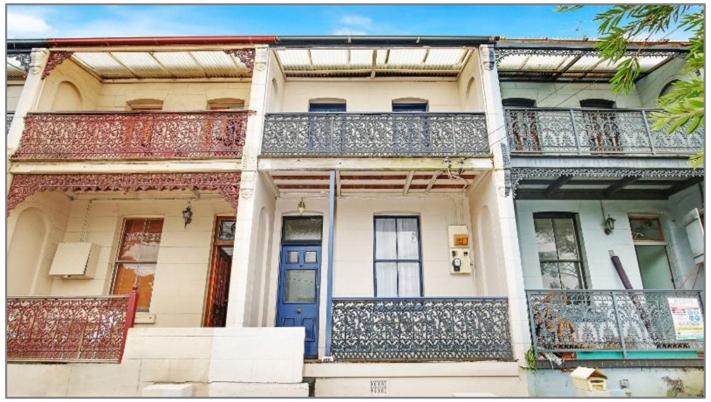
# The Good, the Bad and the Ugly

**… about buying a home**

# What do I mean by a home?

House

Duplex



Terrace

Town House

Apartment or Unit

Caravan, Granny Flat, Tiny House
Retirement Village

# The Good

**Clint Eastwood**
**as ‘Blondie’**

# The Good … about buying a home rather than renting over the long-term

1. **Control over basic needs**

   Provides shelter for your family and also financial and emotional security

2. **Control over dwelling expenses**

   Especially once home loan is paid off … no rent or mortgage payments

3. **Option to make improvements**

   Renovations to suit your needs, tastes, children and/or to improve valuation

4. **Stability in lifestyle**

   No need to worry about a landlord asking you to leave or selling the property

5. **General increase in price**

   Prices are expected to increase over the long-term in line with nominal income growth

# The Good … about buying a home rather than renting over the long-term

6. **Main residence capital gains tax exemption**

   Any gains in your main residence not assessable for income tax

7. **Financial leverage magnifies return on equity**

   If the return on the asset exceeds the interest rate on debt, it magnifies return on equity

8. **Cheap interest rates**

   Residential property is a special form of collateral. Banks are able to lend at low rates.

9. **Tangible**

   Shares and other investments are intangible. It is nice to be able to see your asset

10. **Life gives you some experience**

    A typical 25-year-old can better assess quality of a property than the quality of a share

# The Good … about buying a home rather than renting over the long-term

11. **Forces saving discipline**

    Principal and interest loans forces repayment of loan (saving) over 25 to 30 years

12. **Low levels of market efficiency***

    Opportunities to profit from uninformed or biased buyers or sellers

13. **Excluded from asset tests for social security**

    Main residence is not included in asset tests for social security, such as the Age Pension

14. **Source of funds for retirement**

    Downsizing your home in retirement can free up funds to invest in superannuation

* The concept of 'market efficiency' is concerned with how much current and past information is captured by the current price of an asset. Low levels of market efficiency means that there can be opportunities to profit from uninformed people. But are you informed or uninformed?

# The Bad

**Lee Van Cleef**
**as ‘Angel Eyes’**

# The Bad … about buying a home rather than renting over the long-term

1. **Non-divisibility**

   You cannot buy (or sell) 10% of a residential property. It is binary purchase (lumpy)

2. **Poor liquidity**

   It can take months to buy or sell a property

3. **High minimum investment**

   Entry-level properties are expensive … most people need two incomes

4. **Poor diversification within property sector**

   Most people only have one or two properties

5. **Asset-specific risk**

   Local factors, flooding and leaks, pests (termites), rising damp, concrete cancer, fire …

# The Bad … about buying a home rather than renting over the long-term

6. **Poor diversification across asset categories**

   Cash, fixed interest, residential and commercial property, domestic and internationals shares

7. **High ‘entry fees’ when buying**

   Transfer duty (often 3%+), conveyancing fees, time and hassle (4%+)

8. **High ‘exit fees’ when selling**

   Promotional videos, advertising, agent commissions and conveyancing fees (2%+)

9. **Decreased flexibility**

   High fees and low liquidity makes it harder to move, upgrade or downgrade

10. **Foregone alternatives**

    Investments in other assets such as a diversified portfolio of shares

# The Ugly

**Eli Wallach**
**as ‘Tuco’**

# The Ugly … about buying a home rather than renting over the long-term

1. **Bursting of property price bubbles and recession**

   If property prices fall by 20%+ and you lose your job it can wipe you out

2. **Dodgy property developers**

   Opal Tower, phoenix developers, bait-and-switch, poor construction quality …

3. **Home affordability for new home buyers**

   Too many people buying investment properties contributes to home affordability problems

# So can I just rent my entire life and not buy?

**This called the ‘Rent v Buy Debate’**

# Rent v Buy Debate

## 1. Borrow and buy home

Save $100,000

Borrow $400,000

Buy $500,000 home

Pay loan interest

Pay maintenance costs

Avoid paying rent for the long-term

**Total wealth created over 30 years?**

# Rent v Buy Debate

## 1. Borrow and buy home

Save $100,000

Borrow $400,000

Buy $500,000 home

Pay loan interest

Pay maintenance costs

Avoid paying rent for the long-term

## 2. Rent, borrow and buy investment

Save $100,000

Borrow $400,000

Buy $500,000 investment property

Pay loan interest

Pay maintenance costs

Receive rent and pay rent

**Total wealth created over 30 years?**

# 2. Rent, borrow and invest over long-term

## Advantages

### Greater flexibility with home

Elon Musk just has no home and just rents

### Tax deductions on interest

Interest on investment loan <u>is</u> tax deductible

Interest on home loan <u>not</u> tax deductible

### Diversification reduces risk

Diversified share portfolio

… is better than buying a <u>single</u> property

### Can work for Lean FIRE

Shares generate income, house does <u>not</u>

# 2. Rent, borrow and invest over long-term

## Advantages

**Greater flexibility with home**

Elon Musk just has no home and just rents

**Tax deductions on interest**

Interest on investment loan is tax deductible

Interest on home loan not tax deductible

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Diversified share portfolio

… is better than buying a single property

**Can work for Lean FIRE**

Shares generate income, house does not

## Disadvantages

**Miss out on low interest rates**

Borrowing for shares is more expensive

… than borrowing to buy property

**Life-stages**

Buying home can be attractive with kids

… and very attractive when 60+

**Miss out on forced saving**

Home loan forces you to save

**Speculation**

Risk of gambling on bitcoin!

# Property and the Pie

**Save for peace of mind**

Cash buffer, financial slack

**Save to invest**

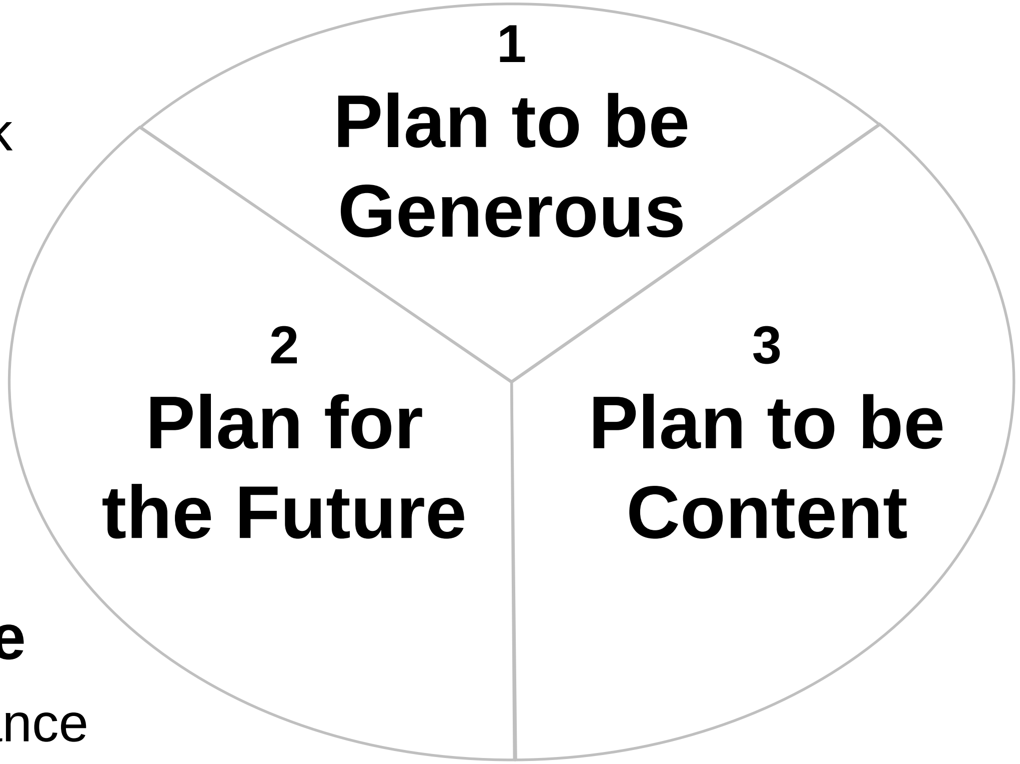
Deposit on property

**Manage debt**

Pay off home loan

**Protect things you value**

Home and contents insurance




**Living expense**

Maintenance costs

Control in long-run

**Leverage and risk**

Fixed v variable costs

# Can International Students buy a home? in Australia

**Normally foreign investors can only buy newly constructed homes.**

**International students may be able to buy new <u>or</u> existing homes.**

Student visa (subclass 500) or Skilled Graduate Visa (subclass 485)

You must have at least 12 months remaining on your visa

**Need permission from Foreign Investment Review Board (FIRB)**

**You must notify the Australian Taxation Office (ATO) of your intent to buy**

**Borrowing here is difficult since you must satisfy financial requirements**

Getting money out of China can be difficult with the USD $50,000 per year limit

**Please do <u>not</u> buy and leave it empty … rent it out!**

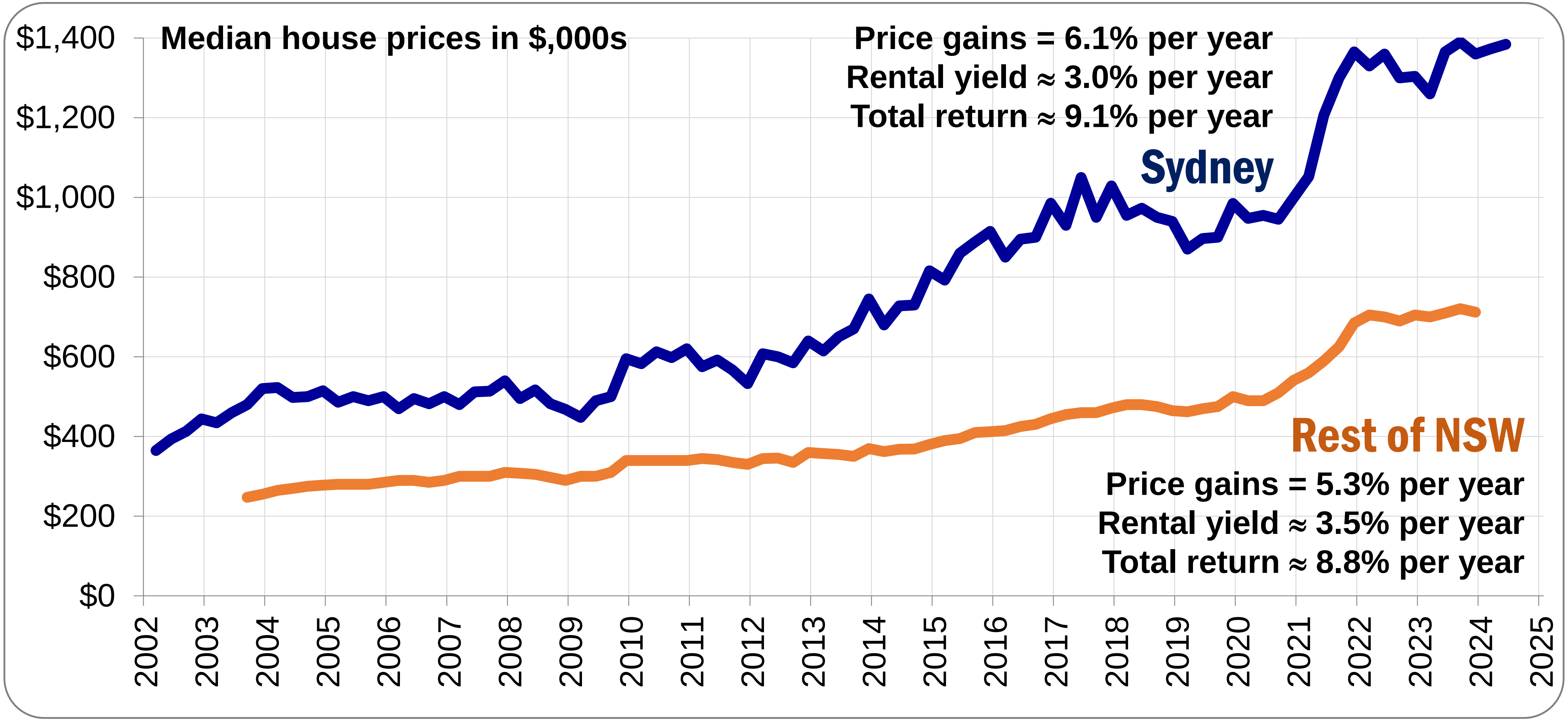

Foreign Investment Review Board (2024) ‘Residential real estate’ at ForeignInvestment.gov.au

Australian Taxation Office (2024) ‘Residential property application for foreign investors’ at ato.gov.au

# Sydney Property Market

**Australia**

# Sydney house prices




Australian Bureau of Statistics (2024) ‘Total Value of Dwellings (Table 2)’ from abs.gov.au

# Property Prices in the short-term

| Demand | Supply |
| --- | --- |
| Number of buyers in the market | Number of sellers in the market |
| Perceived shortages of properties | Perceived glut of properties |
| Fear of missing out (FOMO) | Fear of not getting out (FONGO) |

# Property Prices in the long-term

| Demand | Supply |
|---|---|
| Birth rates | State government land releases |
| Death rates | Local government density rules |
| Migration and overseas students | New home construction |
| Average income | Availability of credit for developers |
| Interest rates | Public transport |
| Live with parents culture | Motorways and roads |
| Savings culture | Technology |
| | **Mainly determined by supply-side** |

# Be careful with house price indices

These are averages (medians)

Micro-markets within Sydney and rural areas can be quite different

The quality of the average dwelling has changed

Median house prices don’t take into account maintenance costs

# Micro-markets

**Prices increase at different rates in different suburbs**

**Determined by supply and demand factors in that area**

**'Gentrification' is one factor that affects demand:**

- When middle class move into areas that were previously lower socio-economic in nature.
- Developers buy properties and develop
- Middle-class renovate their units/homes
- Council spends more money in area.

**If areas have large amounts of land being developed**

**… or large number of new apartments**

**… then additional supply may result in weak property prices**

# Be careful to avoid simply extrapolating …

**Past returns are no indication of future returns!**

# Asset categories real and nominal returns

| | Asset category | Real return | | Inflation | | Nominal return |
|---|---|---|---|---|---|---|
| 1. | Cash | 0.5% | + | 2.5% | = | 3.0% |
| 2. | Fixed Interest | 1.5% | + | 2.5% | = | 4.0% |
| 3. | Residential Property | 5.0% | + | 2.5% | = | 7.5% |
| 4. | Commercial Property | 6.0% | + | 2.5% | = | 8.5% |
| 5. | Australian Shares | 6.0% | + | 2.5% | = | 8.5% |
| 6. | International Shares | 6.0% | + | 2.5% | = | 8.5% |

These are assumed long-term future expected rates of return for this course. The derivation will be discussed further in Units 9 and 10.

# Asset categories price gains and income

| | Asset category | Price gains | | Income | | Nominal return |
|---|---|---|---|---|---|---|
| 1. | Cash | 0.0% | + | 3.0% | = | 3.0% |
| 2. | Fixed Interest | 0.0% | + | 4.0% | = | 4.0% |
| 3. | Residential Property | 5.0% | + | 2.5% | = | 7.5% |
| 4. | Commercial Property | 5.0% | + | 3.5% | = | 8.5% |
| 5. | Australian Shares | 5.0% | + | 3.5% | = | 8.5% |
| 6. | International Shares | 6.5% | + | 2.0% | = | 8.5% |

These are assumed long-term future expected rates of return for this course. The derivation will be discussed further in Units 9 and 10.

# Property Tax

## New South Wales, Australia

# CGT main residence exemption

**Main residence is exempt from capital gains tax (gain in price)**

This must be the case to avoid ‘losing money’ when moving houses

**This makes it an important method for creating and ‘storing’ wealth**

**You are not a foreign resident**

**Is on land of 2 hectares or less**

**It has been your main residence for the entire period**

Note that you are only allowed one main residence at any point in time

**Has not been used to produce assessable income**

Not run a business from it, rented it out or ‘flipped’ it

Australian Taxation Office (2024), ‘Your main residence’ at ato.gov.au

# What about a home office?

**The Australian Taxation Office (ATO) has made a distinction between …**

**A ‘place of business’**

Running and occupancy costs may be an allowable deduction for tax purposes

**CGT main residence exemption is reduced proportionally**

**A ‘place of convenience’**

Only running costs are an allowable deduction for tax purposes

**CGT main residence exemption still applies in full**

**Be careful about claiming tax deductions for working from home**

# What if I rent out a room while I live there?

**Charging rent to a family member is a domestic arrangement**

The rent received is not assessable for income tax

It does not affect CGT main residence exemption

**Charging rent to a ‘border’ for a room in your main residence**

**If you are carrying on a business or renting it out on a platform (like Airbnb)…**

rent received is assessable income and a portion of gain in price may be subject to CGT

**If you are recouping costs on a non-commercial basis …**

rent received is not assessable income and gain in price may not be subject to CGT

**If in doubt … check with an accountant or tax adviser**


Australian Taxation Office (2024) ‘Renting out all or part of your home’ at ato.gov.au

Australian Taxation Office (1985) ‘Taxation Ruling IT2167 Income Tax: rental properties – non-economic rental, holiday home, share of residence, etc. cases, family trust cases’ at ato.gov.au

# CGT exemption for inherited dwellings

**No tax is normally payable on inheritances of main residence**

This must be the case to avoid a perceived ‘death tax’ on main residence

**This makes it an important method of inter-generational wealth transfers**

**Property must be used as a main dwelling by the person inheriting the property**

**… or sold within 2 years of inheritance**

**If the property was used as an ‘investment property’ (rented out)**

**then tax will be payable on some or all of the Capital Gains**

Australian Taxation Office (2024), ‘Inherited property and CGT’ at ato.gov.au

# The ‘6 year rule’

**You can sometimes have a ‘main dwelling’ while not living in it**

**You can treat the dwelling as you main residence for:**

1. **Up to six years if it is used to produce income (usually rent)**
2. **Indefinitely if it <u>not</u> used to produce income**

**You cannot treat any other dwelling as main residence during the same period**

**It does not apply to a period before it became your main residence**

**It does not apply to properties owned by a company or trust**

**You can ‘reset’ the 6 years by moving back into it (with evidence)**

Australian Taxation Office (2024), ‘<u>Treating a former home as main residence</u>’ at ato.gov.au

# NSW Stamp Duty also known as ‘transfer duty’

## NSW State Government tax on transfer of ownership of property

Considering allowing people to choose between upfront stamp duty or a regular land tax

| Property value | Transfer duty |
|---|---|
| **$0 - $16,000** | $1.25 for every $100 (minimum $10) |
| **$16,000 - $35,000** | $200 plus $1.50 for every $100 over $16,000 |
| **$35,000 - $93,000** | $485 plus $1.75 for every $100 over $35,000 |
| **$93,000 - $351,000** | $1,500 plus $3.50 for every $100 over $93,000 |
| **$351,000 - $1,168,000** | $10,530 plus $4.50 for every $100 over $351,000 |
| **$1,168,000 - $3,505,000** | $47,295 plus $5.50 for every $100 over $1,168,000 |
| **$3,505,000 or above** | $175,830 plus $7.00 for every $100 over $3,505,000 |

NSW Government Revenue (2024), ‘Transfer duty’ at revenue.nsw.gov.au

# Lenders Mortgage Insurance (LMI)

## Not really a 'tax' but a fee if you borrow more than 80% of value of property

… used to 'insure' the banking system against a sudden collapse in property prices

| Property ($) | Deposit ($) | Loan ($) | Deposit (%) | Loan (%) | LMI ($) | LMI (%) |
|---|---|---|---|---|---|---|
| 1,000,000 | 200,000 | 800,000 | 20% | 80% | - | - |
| 1,000,000 | 150,000 | 850,000 | 15% | 85% | 11,305 | 1.13% |
| 1,000,000 | 100,000 | 900,000 | 10% | 90% | 22,410 | 2.24% |
| 1,000,000 | 50,000 | 950,000 | 5% | 95% | 43,605 | 4.36% |

Property value here is assumed to be the total cost of buying the property including property price, transfer duty and other costs
LVR is a loan to value ratio and is calculated as Loan / Property value (including transfer duty and costs)
LMI is sometimes waived for doctors, accountants, lawyers, mining specialists, professional athletes (or entertainers) and for DHA properties.
LMI can be waived for first home buyers with 5% deposit under First Home Buyer Guarantee (FHBG)
Australian Government (2024), 'First Home Guarantee' at nhfic.gov.au

# NSW Land Tax

**A state tax payable on investment properties**

**Primary residence is generally exempt**

- It must have been your primary dwelling for the entire year
- You must own the property directly (not via company or trust)

**Based on assessed value of 'land'**

- Which is usually less than the value of land + building

**Land tax free threshold $1,075,000 (2024)**

- Starts at 1.6% + $100 of combined land value above threshold
- Premium threshold of 2% cuts in at $6,571,000 (2024)

**A common strategy is to hold one investment property in each state**

NSW Government Revenue (2024), 'Land tax' at revenue.nsw.gov.au

# Goods and Services Tax (GST)

**Payable on newly constructed homes or apartments**

It is <u>not</u> payable when buying an existing residential property

**10% GST payable on building component (not land)**

**10% GST is payable on agent commissions when selling any property**

… but is <u>not</u> payable on the rent collected from tenants on a residential property

# Income tax on investment properties

**Rent received is assessable income and subject to income tax**

**Gain on sale of property is subject to income tax (capital gains tax)**

- Only 50% of the gain is assessable if held for more than one year

**Allowable deductions include:**

- Interest expenses on investment loan
- Body corporate fees, council rates and water charges
- Land tax
- Agent fees, advertising for tenants, repairs, maintenance and cleaning
- Insurance

**… make sure that you use an accountant to minimise errors**

Australian Taxation Office (2024), 'Rental property expenses to claim' at ato.gov.au
Australian Taxation Office (2024), 'Rental expenses you can claim now' at ato.gov.au

# Property Valuation

# The value of a residential property

| | |
|---|---|
| Non-monetary benefits | Garden, pool, location … |
| Option to make improvements | Subdivide, renovations … |
| Present value of tax benefits | Decreases required return |
| Present value of future rent | PV of Perpetuity |

# Valuation approaches

1. **Relative Prices**
2. **Price-Earnings Multiple**
3. **Value per square metre**
4. **Statistical regression**
5. **Present value of future rent**
6. **Option to make improvements**

**Comparables** (1–4)

Assumes observed prices are ‘fair’ and not in property asset price bubble

**Discounted cash flow** (5)

**Real option analysis** (6)

See my Advanced Finance channel
… but it is very very tough

# 1. Relative prices

**Observe ask and sale prices of similar properties**

- Asking price is available on domain.com.au and realestate.com.au
- Actual sale price is available on homepriceguide.com or by asking agent
- Observe percentage difference between ask and final sale price
- Observe time taken from first listing to final sale

**Track asking price against actual in a spreadsheet**

An underlying assumption of this method is that the observed prices are indeed ‘fair’ and there is no property asset price bubble

# 2. Price-Earnings Multiple

1. **Observe sale price of properties sold (P)**
2. **Observe asking weekly rental prices for similar properties (C)**
3. **Estimate gross rent collected over one year ($C \times 50$)**
   50 weeks assumes that 2 weeks per year will be untenanted
4. **Calculate Price-Earnings multiple**
   $$\mathbf{PE} = \frac{\mathbf{P}}{\mathbf{C} \times \mathbf{50}}$$
5. **For any new property, estimate annual gross rent ($C \times 50$)**
6. **Multiply by PE for similar properties to estimate market value**

An underlying assumption of this method is that the observed prices are indeed ‘fair’ and there is no property asset price bubble

# 3. Value per square metre

**This approach works best for apartments since they have no land**

1. **Observe sale price of apartment sold (P)**
2. **Find out the number of square metres of floor space (N)**
3. **Calculate value per square metre (V)** $$\mathbf{V = \frac{P}{N}}$$
4. **For similar apartment, ask agent number of square metres (N)**
5. **Multiply V × N to estimate market value of apartment (P)**

The approach can work for houses but you need to do it for land and building separately

An underlying assumption of this method is that the observed prices are indeed 'fair' and there is no property asset price bubble

# 4. Statistical regression

**The approach taken by professional property price estimation companies**

**Collect prices and property features for many different properties**

**Estimate parameters (β) for relationship between variables (X) and price (P)**

$$\mathbf{P_i} = \beta_{\mathbf{0}} + \beta_{\mathbf{1}}\mathbf{X_1} + \beta_{\mathbf{2}}\mathbf{X_2} + \beta_{\mathbf{3}}\mathbf{X_3} + \beta_{\mathbf{4}}\mathbf{X_4} + \beta_{\mathbf{5}}\mathbf{X_5} + \beta_{\mathbf{6}}\mathbf{X_6} + \boldsymbol{\varepsilon}_{\mathbf{i}}$$

An underlying assumption of this method is that the observed prices are indeed ‘fair’ and there is no property asset price bubble

# 5. Present value of future rent in perpetuity

| 0 | 1 | 2 | 3 | 4 | … |
|---|---|---|---|---|---|
| | $C_1$ | $C_1(1+g)^1$ | $C_1(1+g)^2$ | $C_1(1+g)^3$ | … |

**P**

$$P_{\infty,g} = \frac{C_1}{r - g}$$

**'$C_1$' is the annual net rent (after agent and other costs)**

**… growing at a rate of 'g' per year forever ∞**

**'r' is the required rate of return per period** (based on investments with similar risk)

**'P' is value today of the property that can generate that rent**

This is the first approach that we have covered that uses valuation fundamentals and doesn't assume current prices are 'fair'

# 5. Present value of future rent example

**How much would John be willing to pay for an investment property today if he believes it will generate $20,000 per year in rent indefinitely (after deducting agent fees and other costs), growing at 2.5% above inflation if he can achieve a real rate of return on similar investments of 5% per annum?**

$$\mathbf{P_{\infty,g} = \frac{C_1}{r - g}}$$

This is the first approach that we have covered that uses valuation fundamentals and doesn't assume current prices are 'fair'

# 5. Present value of future rent example

**How much would John be willing to pay for an investment property today if he believes it will generate $20,000 per year in rent indefinitely (after deducting agent fees and other costs), growing at 2.5% above inflation if he can achieve a real rate of return on similar investments of 5% per annum?**

$$\mathbf{P_{\infty,g} = \frac{C_1}{r - g}}$$

$$\mathbf{= \frac{20,000}{0.05 - 0.025}}$$

$$\mathbf{= \$800,000}$$

This is the first approach that we have covered that uses valuation fundamentals and doesn't assume current prices are 'fair'

# 6. Option to make improvements

**Buying the property gives you the right, but not the obligation**

**… to make a follow-on investment (renovation)**

**… that could significantly increase the value of the property**

**In finance, this is called a ‘real option’ and it is valuable.**

**This explains why houses are more valuable than apartments**

- There are more ‘real options’ to make improvements to land + building (in perpetuity)
- … than there are to just make improvements to an apartment

Real Option Analysis is covered in my Advanced Finance channel … but it is very challenging!

# Human Bias and Property

# Buyers are biased by comparisons

| | |
|---|---|
| 1. Cheap property | $500,000 |
| 2. Expensive property | $800,000 |

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# Buyers are biased by comparisons

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| **1. Cheap property** | **$500,000** |
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| **3. Complete rip-off** | **$1,000,000** |

Same as expensive property + something not valuable

# Buyers are biased by comparisons

**1. Cheap property** **$500,000**

**2. Expensive property** **$800,000** ☑

**3. Complete rip-off** **$1,000,000**

Same as expensive property + something not valuable

# Human bias when buying

## Halo effect

Attractiveness of the real-estate agent makes the house more trustworthy

Sequencing of properties from ‘rip-off’ to ‘expensive’ makes ‘expensive’ seem ‘good value’

## Anchoring bias

Anchored on first price and adjust too little based on new information

## Affect bias

Falling ‘in love’ with small features makes you willing to pay a premium

Visualising ‘having children’ or ‘growing old’ in the property

## Endowment effect

Visiting the property in-person and touching surfaces

… makes you feel a sense of ownership that inflates perceived value

# Human bias when selling

## Endowment effect

You value something more highly when you own it

… and made worse by time and effort invested into renovations

## Egocentric bias

Think that you have ‘good taste’ and others share your taste

## Affect bias

Cherished memories and emotional attachment

## Disposition effect

Reluctance to sell at a loss

# Human bias and auctions

## Reciprocity effect

Free coffee and ice-cream

## Perceived scarcity

Opening pitch

## Anchoring and adjustment bias

Suggest a high reference price

## Consistency effect

Start from below reservation price to get people to keep bidding

## Loss aversion

Creating the fear of missing out

# Winner’s curse

**Derived from ‘game theory’ for auctions in Economics**

**The winner in a competitive auction overpays**

You paid more than the maximum value attributed to the property by everyone else

1. **Be very clear about your maximum price**
2. **Be willing to lose**
3. **Avoid getting into a ‘bidding war’**
4. **Make an offer the week before the auction (with deadline)**
5. **Make an offer the week after a property is ‘passed in’**
6. **Consider using a professional property buyer’s agent**

# Property market inefficiency

**There are no sophisticated 'institutional' investors**

**There are some professional buyer's agents**

**Most individual buyers and sellers are relatively inexperienced**

- Failure to identify valuable real options to make improvements or renovations
- Non-financial motives
- Poor liquidity and high transaction costs
- Unique and non-divisible nature of individual properties
- High costs of 'bridging finance' for people who have bought without selling
- Urgent sales from deceased estates or insolvencies

**These market inefficiencies can provide opportunities**

# Before you buy

## New South Wales, Australia

# Process

1. **Preliminary loan and property research (3 months)**
2. **Arrange pre-approval for a home loan (allow 1 month)**
3. **Active property search and price negotiations (3 months +)**
4. **Exchange of contracts (1 day)**
5. **Settlement (6 weeks)**
6. **Move in (1 day)**
7. **Renovations (often several in the first year or two)**

# Before you look …

**Talk to your bank to determine how much they will lend**

**Determine some realistic locations and dwelling type**

- Likely need to go a long way SW or NW for affordability
- Consider proximity to transport, work, family …
- Consider starting with a two bedroom apartment (renting out the second room)

**Research property prices**

- Domain.com.au and realestate.com.au

# Apartment v houses

| House | Apartments |
|---|---|
| Own the land and building | Own ‘air’ between the walls |
| Option to make major improvements | Option to make minor improvements |
| Higher price gains over long-term | Higher rental yield (rent / price) |
| Higher maintenance costs | Lower maintenance costs |
| No body corporate fees | Body corporate fees |
| Very high starting price | High starting price |

# Torrens v Strata Title

**Torrens title (houses)**

The purchaser owns the land and building … forever

**Strata Title (apartments)**

Multiple property owners on one piece of land with shared responsibility for common areas

**Leasehold title**

Government owns the land and you have a lease for X years (common in other countries)

**Community title**

Property is shared and owned by many people with fees for common 'community areas'

**Retirement villages**

Can be strata, leasehold, community or other types … get legal advice it can be shonky!

Firstmac (2018) 'Understanding Different Types of Property Titles in Australia' at firstmac.com.au

# Consider ownership structure

1. **Individual**
2. **Joint Tenancy**

   **Buying the property in unity with other parties (usually a spouse)**

   The title automatically transfers to other owners upon your death

   … bypassing your will
3. **Tenancy in Common** (or Tenants in Common)

   **Allows different owners to have a claim over specific percentage of value**

   The title does not automatically transfer to other owners upon your death

   … and is subject to instructions in your will
4. **Trust or company**

   A sophisticate strategy for ‘asset protection’ but may be subject to more land tax

# Getting the finances ready

**Banks are often willing to lend up to 95% of the property value**

**This is generally a really bad idea**

- Lenders Mortgage Insurance is murder (more than 4%)

**Best to only borrow 80% of the property value if possible**

- … I acknowledging that this can be tough for first-home buyers

**Go super-lean and work hard to save a deposit of 20%**

- Later we will consider using a ‘Family Security Guarantee’ (use parents as guarantor)
- … this will allow you to borrow up to 100% and then immediately pay off 20% of the loan

# First home-owner assistance in NSW

**You (or spouse) cannot have owned a home previously**

You must move into it within 12 months and live in it for continuous 12 months

**First Home Buyers Assistance Scheme (FHBAS)**

Zero stamp duty (transfer duty) on existing homes up to $1 million (saves up to ~$40,000)

**First Home Owner Grant (New Homes)**

Additional $10,000 toward purchase of land and home up to $750,000 in value

**First Home Super Saver Scheme (FHSS)**

Withdraw $50,000 on voluntary super contributions to buy home (double for couples)


NSW Government (2024) ‘First Home Buyer’ at nsw.gov.au
Australian Taxation Office (2024), ‘First home super saver scheme’ at ato.gov.au
Australian Government (2024), ‘First Home Guarantee’ at nhfic.gov.au

# First home-owner assistance in NSW … continued

## First Home Buyers Guarantee (FHBG)

No Lenders Mortgage Insurance (LMI) if buying a home with only 5% deposit.

## Shared Equity Home Buyer Helper

First-home buying Single parents, Over 50s and Key Workers

Nurses, Midwives, Paramedics, Teachers, Early Childhood Educators and Police Officers

NSW Government contributes up to 30% of equity on existing homes

NSW Government (2024) ‘First Home Buyer’ at nsw.gov.au
Australian Taxation Office (2024), ‘First home super saver scheme’ at ato.gov.au
Australian Government (2024), ‘First Home Guarantee’ at nhfic.gov.au

# Arrange pre-approval on your loan

- **Preferably do this before making any offers**
- **Decide on your financial institution and specific home loan product**
- **Make a written application involving application forms**
  - Apply for the maximum amount possible
  - Provide information on your income, expenses, assets and liabilities
  - Provide evidence of income and regular saving
- **You should obtain written ‘pre-approval’**
  - Subject to them being satisfied with the property that you buy is adequate security for loan
  - Usually given within a week of receiving paperwork
- **They rarely ‘reject’ a loan after written pre-approval is given**

# Property searching tips

1. **Be flexible**

   You probably can’t afford the suburb you want

2. **Use property websites**

   Domain.com.au and realestate.com.au

3. **Take a virtual stroll down the street**

   Using Google Street View

4. **Research actual sale prices**

   … or properties in the street or building using homepriceguide.com.au

5. **Get to know the local agents**

   Be courteous and professional … and learn their names

# Property searching tips

6. **Rent in the area before buying**

   Inspecting properties is much easier and you get to know the area

7. **Jog in the area and get to know the streets**

   Save on gym fees and get to know the local area

8. **Remember that the agent is acting for the vendor (seller)**

   Don’t let them know that your heart is set on the property and will pay any price!

9. **Estimate the ‘fair market value’**

   Relative price, P-E multiple, Value per square metre …

10. **Be patient, avoid emotional attachment and human bias**

# Traditional tips on location

**The golden rule is “location … location … location!”**

1. **Closer to the city centre the better**
   Although this could change with the trend to work-from-home
2. **Close to transport and health services**
3. **Where gentrification is happening or will happen soon**
4. **Is aesthetically pleasing or has the potential to be so**
5. **Has a unique or rare positive characteristic**
   Scarcity strengthens your negotiating position when selling

**… but everyone knows these tips!**

# My tips on location

**Consider both supply and demand of properties**

**Supply is driven by new home construction**

**Demand is driven by people wanting to live there**

**Slow growth in supply and strong growth demand is good**

Limited new land releases or new apartment construction

**Strong growth in supply is often bad (independent of demand)**

# Break

# Making an offer

**New South Wales, Australia**

# Before you make an offer …

## Building and pest inspection

An expert looks for damage, structural problems or pest related issues

Typical cost is somewhere between $300 to $500

## Strata search (for apartments)

An expert looks for problems with apartment building or management

Typical cost is somewhere between $250 and $400

## You can also make a ‘contingent-offer’ on the results of the inspection

# Conveyancer or solicitor

## Who will handle all the 'due diligence' and 'legal paperwork'?

1. **Do it yourself**

   Not recommended as it is specialised work

2. **Solicitor**

   Many solicitors (lawyers) handle property conveyancing as one of many matters

   More expensive and may be better if an unusual situation arises

   Tend to pass-on much of the work to legal secretary who does <u>not</u> have legal qualifications

3. **Conveyancer**

   Property conveyancers do not have legal qualifications but <u>only</u> do property conveyancing

   Less expensive and very experienced but may miss problems in unusual cases

   Quality and capability of property conveyancers can vary significantly

   A good starting point is conveyancing.com.au

# Making an offer

## There is an ‘art’ to negotiation

- Find out the rules (eg. Low first offer, disinterest, be professional and courteous)
- Practise your skills on some properties that you don’t want … but don’t be a time-waster
- Remember that the ‘nice’ agent is acting on behalf of the vendor (seller)

## Tell your bank and conveyancer that you are making a serious offer

- They <u>both</u> may ask for a copy of the contract of sale (available from the real-estate agent)

## Put an offer to the agent

- They will pass the offer on to the vendor (seller)
- There will usually be some ‘back and forth’ as they ask for more
- … or wait for a better offer from someone else

# Exchanging contacts

**In Australia, a verbal acceptance of an offer is worthless**

The vendor can still 'shop around' and contact other buyers asking for a higher offer

If another buyer offers them more … consider yourself 'gazumped'

1. **Make sure your conveyancer or solicitor has reviewed contract of sale**
2. **Arrange for 10% deposit**
3. **Go to the real-estate agent immediately with signed contact of sale**
4. **Exchange contracts and pay the deposit**

   0.25% of purchase price penalty if withdraw within the 5-day cooling-off period

   Vendor may ask for a Section 66W Certificate to waive your 5-day cooling-off period

Department of Fair Trading (2024), 'Contract and deposits' at fairtrading.nsw.gov.au

# After exchanging contracts …

**You have 6 weeks (42 days) until settlement and the clock is ticking …**

Settlement is when the remaining 90% of funds will need to be paid

**Tell your bank that you have exchanged contracts**

Sign all necessary home loan documentation

Best to get legal advice about home loan contract before signing it

**Solicitor or conveyancer does all the other ‘conveyancing’ work**

Best to ‘follow them up’ to make sure everything is going smoothly

**Book furniture removalists, friends and family**

Furniture removalists often need to be booked about 6 weeks in advance

**Arrange house insurance, electricity, water and broadband connection**

The bank will require evidence that you have insurance

# Settlement

**Settlement involves the final 90% payment and transfer of title**

**Standard is exactly 6 weeks (42 says) after exchanging contracts**

- Some vendors may accept a delayed settlement period

**Do a final inspection on the morning of settlement day**

- Visit the property and make sure it hasn't been damaged by tenants (or burnt down)

**You do <u>not</u> attend the settlement**

- Settlement is done electronically between banks and solicitors or conveyancer for each party
- They check that paperwork, transfer of title and finances are all in order

**If successful, your solicitor or conveyancer will notify you**

- … if not successful, they will notify you of the reason for failure and the new date

Department of Fair Trading (2024), '<u>Contract and deposits</u>' at fairtrading.nsw.gov.au

# After settlement

**Contact electricity, water and broadband to confirm connection**

**Pick up the keys from the real-estate agent**

**Move in**

**Notify others of change of address**

- Driver’s licence with Roads & Maritime Services
- Australian Electoral Commission
- Employer
- Financial institutions (bank, super funds, investments)

**Throw a house-warming party with friends!**

# After You Buy

# On-going ownership costs

Council rates

Water and sewerage service

Maintenance costs

Body corporate and strata management fees (apartments)

# Renovating

**Most people invest too much in renovations**

**If price goes up … how much is from renovations?**

**Avoid being the best house in the street**

**Maintenance important to keep the value**

- Pest inspections, guttering, external painting etc.

**Avoid spending more than 10% of purchase price**

**Renovating is expensive and may not add as much value as you think**

- Watch out for the renovation hype

# Renovating an old two bedroom apartment

1. **Learn to paint**
   Use good quality paint and special paint for bathrooms
2. **Kitchen**
   Replace kitchen cupboard doors, taps (quality), cooktop and oven
3. **Bathroom**
   Re-grout tiles (professional), replace mirror, cabinet doors and taps
4. **Lights**
   Replace old light fixtures and but lots of modern lamps
5. **Flooring**
   Replace carpet or polish floorboards
6. **Technology**
   Invest in high quality wireless router and repeaters. Add Alexa or Google Home

# Renovations that may <u>not</u> add value

**Expensive professional landscaping**

**Unusual colour schemes**

**Renovations that do not have council approval**

**Pool**

# Selling … and then buying another property

**The timing of selling a property then buying another is difficult**

Uncertain time to find property to buy

\+ uncertain time to sell current property

**Sometimes ‘emotional’ people buy before they sell old property**

**This is usually a big mistake**

**Bridging finance is expensive and difficult to obtain**

# An alternative strategy for moving house …

1. Decide where you would like to live next
2. Rent a home in the new location
3. Rent out your old home then put it on the market for sale
4. Wait until your old home is sold
5. Search and buy your new home at leisure

# Property and divorce

**Divorce can be emotionally and financially devastating**

Marriage and de facto relationships are normally treated the same

**The default situation is that assets are split evenly**

This can depend on whether you signed a Binding Financial Agreement (prenup)

Family Court will favour the parent who is primary carer for children

1. **Make sure you have been together for 5 years before buying together**
2. **Men should expect zero equity upon family break-down**

The main residence is usually awarded to the mother (if there are dependent children)

The father is still expected to support the children financially

… but now has to pay rent to live somewhere else

This is a generally a good system because it protects the vulnerable

# 20 Property Strategies

# 1. Buy and Hold

**Buy investment property and hold it for the very long-term (20+ years)**

**Never pay tax on capital gains because you don't sell it**

**Use equity from property to buy other investment properties or shares**

**Gradually moves from negative to positive cash flow**

- Repayments remain roughly the same over time
- Rent gradually increases in line with nominal average incomes (including inflation)

# 2. CGT Main Residence Exemption

**Main residence is one of the few assets**

**… with non-assessable gains for income tax**

**Make good use of it!**

**For couples with two properties**

**... each individual can have different declared ‘main residence’**

# 3. Negative Gearing

1. **Buy investment property**
2. **Take out maximum investment loan possible as interest-only loan**
3. **Make sure your investment property reduces taxable income**

   Assessable income: Rent

   Allowable deductions: Investment loan interest, agent fees, maintenance ….

   Taxable income = Assessable income less allowable deduction
4. **Offset 'loss' against salary income to reduce income tax**
5. **Hold property for long-term to achieve gain in asset value**

**If you do this while renting elsewhere it is called 'rentvesting'**

Rentvesting gives you flexibility to rent where you need to live and investing elsewhere

# 4. Positive Cash Flow

1. **Build up portfolio of investment properties**
2. **Focus on properties that generate good rental income**

   These may be lower valued properties in areas of high rental demand
3. **Make sure portfolio generates positive cash flow**

   Cash inflow: Rent

   Cash outflows: Investment loan interest, agent fees, maintenance ….
4. **Stop working and enjoy ‘passive income’ (financial independence)**

# 5. Lean Boomerang

1. **Move out of home … enjoy some independence and save nothing!**
2. **Move back into home with parents**
3. **Extreme saving (70% of income) … to save deposit on home**

**More people are asking for help from family these days** (Bank of Mum & Dad)

1. **Parents give $100,000 as a gift**
2. **Parents give $100,000 as a low interest rate loan**
3. **Parents invest $100,000 for a portion of the title of the property**

# 6. Investor Boomerang

1. **Buy two bedroom apartment and live in it for 6 months**
2. **Move back home with parents**
3. **Rent out two bedroom apartment as investment property**
4. **Keep 2 bedroom apartment as your ‘main residence’**

   Use 6 year rule to maintain CGT main residence exemption

# 7. Trend-setter

**The trendy inner-city suburbs are all completely unaffordable**

**‘Walk your own path’ further out in NW, W and SW Sydney**

# 8. Buy New or Off-The-Plan

**Not recommended**

**There can be significant time delays for construction completion**

**The market may have ‘turned’ in by the time property is completed**

**There have been a lot of problems with phoenix developers**

**… and poor quality construction**

**There is often a ‘premium’ paid on new homes**

**Buy apartments that are 20+ years old because problems have surfaced**

# 9. Distressed Sales

Deceased estates

Insolvencies

People who have bought new property but cannot sell old one

Offer distressed sellers a quick transaction and settlement

These are ‘vulture’ acquisitions

# 10. Renovate to Hold

1. **Buy undervalued property that needs renovation**
2. **Renovate them yourself**
3. **Rent out the property**

   Some couples ‘tag-team’ their CGT main residence

**It is very easy for renovations to go over budget**

**You need to make renovations that will attract good tenants**

**… and not overcapitalise**

# 11. Renovate to Flip

1. **Buy undervalued property that needs renovation**
2. **Renovate them yourself**
3. **Sell property for a profit for 'tax-free income'**
   Usually people claim CGT main residence exemption so don't include gain in tax return
   Some couples 'tag-team' their main residence so you don't have to live in construction zone!

**ATO has made it clear that gain from flipping is assessable income**

**Works best in a 'hot' property market**

How much of the gain is from the renovation and how much from general price increase?

**Possibly an effective strategy for young tradespeople**

Interest on home loan and stamp duty can really cut your margins

Alasdair Duncan (2023), 'House Flipping in Australia – Easy Profit or Overhyped Fantasy?' at canstar.com.au

# 12. Granny Flat

**Buy a house with a larger block of land (usually above 500m$^2$)**

**Build a small 'granny flat' on a property for under $80,000**

**If it complies with the law, council approval may not be necessary**

**Rent it out for more than $200+ per week**

**This is a gross rental yield of $200 × 50 weeks / $80,000 = 12.5%**

State Environment Planning Policy (Affordable Rental Housing) 2009
NSW Government (2024) 'Complying Development: Granny Flats' at planningportal.nsw.gov.au

# 13. Dual Occupancy

Don’t just build a granny flat … build an entire second dwelling

Earn a second income to help pay your mortgage payments

… but tends to be a lower rental yield than the ‘granny flat’ option

Requires council approval

Council rates will also increase

# 14. Springboard

**Don’t buy your dream-home as your first home**

It will take ages to save the deposit and you will have a huge home loan

1. **Buy a two bedroom apartment that is at least 30 years old**

   The quality of newly constructed apartments is highly questionable

   … there are a lot of shonky operators and ‘phoenix’ developers

   Most problems with an apartment building have surfaced after 30 years

   There are also opportunities to do some internal renovation yourself ‘on the cheap’

2. **Rent out the second bedroom to a border**
3. **Aggressively pay down the home loan and wait for price increase**
4. **Buy dream-home while retaining apartment as investment property**

# 15. Tag-Team

**For stable couples who have made life-long commitments …**

1. **Buy Property A under name of Person 1 (highest income)**

   Live in Property A for a while as main residence of Person 1 (with evidence)

2. **Later … buy Property B under name of Person 2 (lowest income)**

   Rent out Property A as investment property

   Borrow maximum investment loan against Property A

   Person 1 claims interest as tax deduction (rent is assessable income)

   Person 1 may even be able to continue to declare Property A to be ‘main residence’ under the 6-year rule for CGT relief for up to 6 years

3. **Focus on paying off home loan for Property B**

   Property B is main residence for Person 2

   Interest on home loan is not tax deductible so pay it off as soon as possible

# 16. Full-House

**If you are life-long friends with another couple with no kids**

1. **Buy one small house owned by four adults**

   Purchase as ‘tenants in common’ with specific ownership percentages

   Use a solicitor to draft an agreement to cover contingencies

   Each couple has separate home loan (preferably with the same bank)

   Save on living costs shared amongst four adults

2. **Aggressively pay down home loan with four incomes for X years**

3. **After X years both leave and retain as investment property**

   Both couples borrow against equity to buy their next property

# 17. Inter-State Roll-Up

1. **Buy first property in Sydney**

   Aggressively pay down home loan and wait for increase in price

2. **Buy second property in Melbourne**

   Borrow using equity from first property and second property as collateral on loan

3. **Buy third property in Brisbane**

   Borrowing using equity from first two properties and third property as collateral

4. **Buy fourth property in Perth**

   Borrowing using equity from first three properties and fourth property as collateral

5. **…**

Buying properties in different states helps to avoid paying land tax

Is sometimes done in ‘trust’ structure but this can have implications for land tax in NSW

# 18. Subdivision

**Buy a property with a large land size (usually above 1,000$m^2$)**

**Wait for local council to change subdivision rules**

**Sub-divide property into two then:**

1. **Sell off 500$m^2$ parcel of land; OR**
2. **Build a house and sell land and house**

# 19. Downsizing

**After age 65 …**

1. **Kick out any remaining adult kids (boomerangs)**
2. **Sell the 4 bedroom family home that you have owned for 10+ years**
3. **Buy a smaller 2 to 3 bedroom home or 2 bedroom apartment**
4. **Invest $300,000 each into superannuation**

   Note that the Downsizing Contribution into super can be done as early as age 60

# 20. Commercial Property

Shops, small offices or warehouses

Stable tenants over many years (reduces risk)

Higher rental yield than residential property

Diversification away from residential property reduces some risk

… but what if businesses leave the area (increases risk)

… and minimum entry price can be a lot higher than residential property

# Remember … The Bad

1. **Non-divisibility**

   You cannot buy (or sell) 10% of a residential property. It is binary purchase (lumpy)

2. **Poor liquidity**

   It can take months to buy or sell a property

3. **High minimum investment**

   Entry-level properties are expensive … most people need two incomes

4. **Poor diversification within property sector**

   Most people only have one or two properties

5. **Asset-specific risk**

   Local factors, flooding and leaks, pests (termites), rising damp, concrete cancer, fire …

# Remember … The Bad

6. **Poor diversification across asset categories**

   Cash, fixed interest, residential and commercial property, domestic and internationals shares

7. **High ‘entry fees’ when buying**

   Transfer duty (often 3%+), conveyancing fees, time and hassle (4%+)

8. **High ‘exit fees’ when selling**

   Promotional videos, advertising, agent commissions and conveyancing fees (2%+)

9. **Decreased flexibility**

   High fees and low liquidity makes it harder to move, upgrade or downgrade

10. **Foregone alternatives**

    Investments in other assets such as a diversified portfolio of shares

# … and the Ugly

1. **Bursting of property price bubbles and recession**

   If property prices fall by 20%+ and you lose your job it can wipe you out

2. **Dodgy property developers**

   Opal Tower, phoenix developers, bait-and-switch, poor construction quality …

3. **Home affordability for new home buyers**

   Too many people buying investment properties contributes to home affordability problems

# Discussion

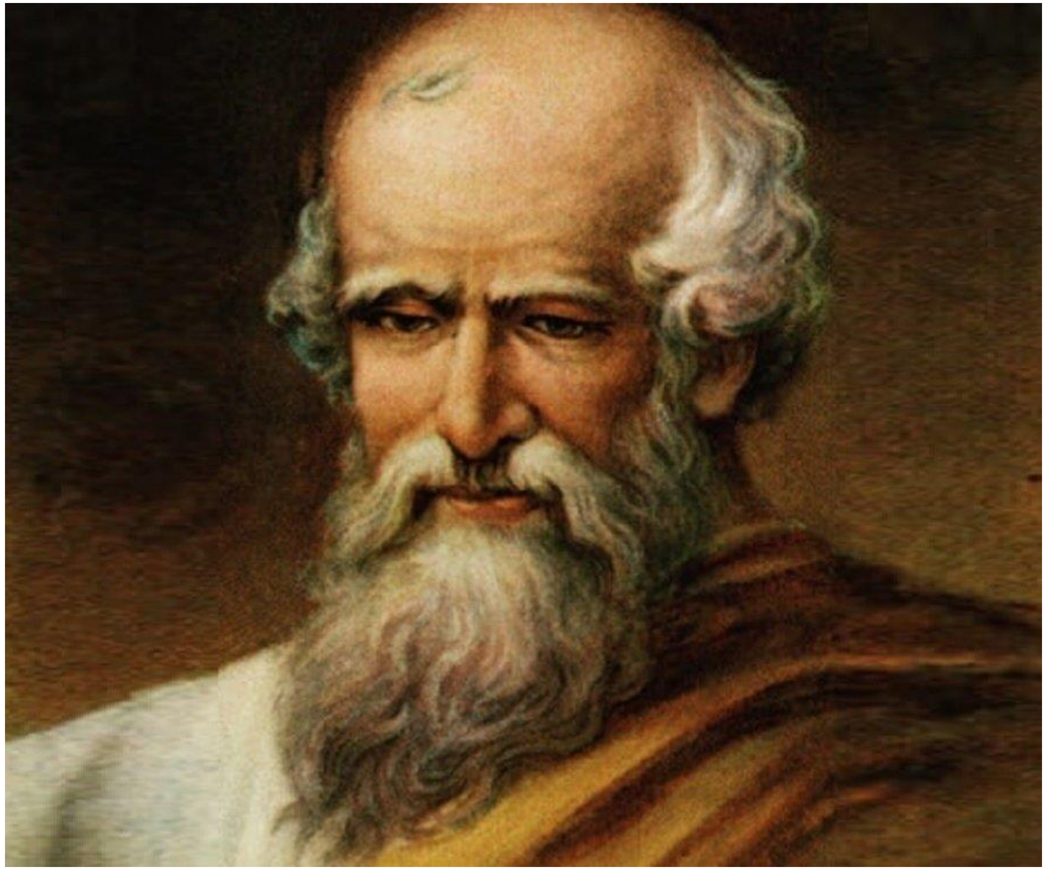
## Are there any strategies we covered that sound interesting to you?

## Is there a strategy that we missed?

1. Buy and Hold
2. CGT Main Residence Exemption
3. Negative Gearing
4. Positive Cash Flow
5. Lean Boomerang
6. Investor Boomerang
7. Trend-setter
8. Buy New or Off-The-Plan
9. Distressed Sales
10. Renovate and Hold
11. Renovate and Flip
12. Granny Flat
13. Dual Occupancy
14. Springboard
15. Tag Team
16. Full House
17. Inter-State Roll-Up
18. Subdivision
19. Downsizing
20. Commercial Property

# Financial Leverage

# Archimedes



Famous mathematician and inventor

in ancient Greece (3rd century B.C.)

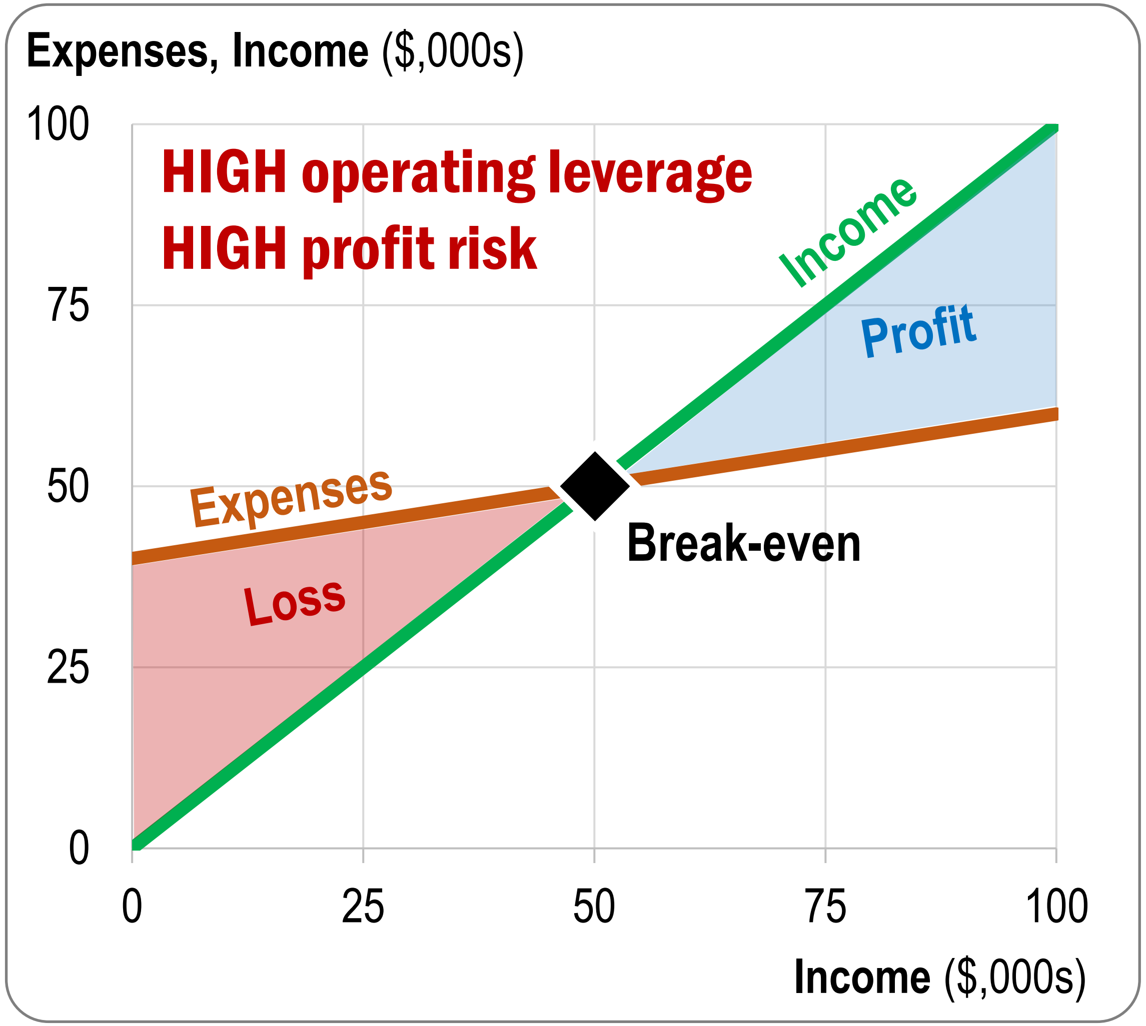
"Give me a lever long enough

and a fulcrum on which to place it

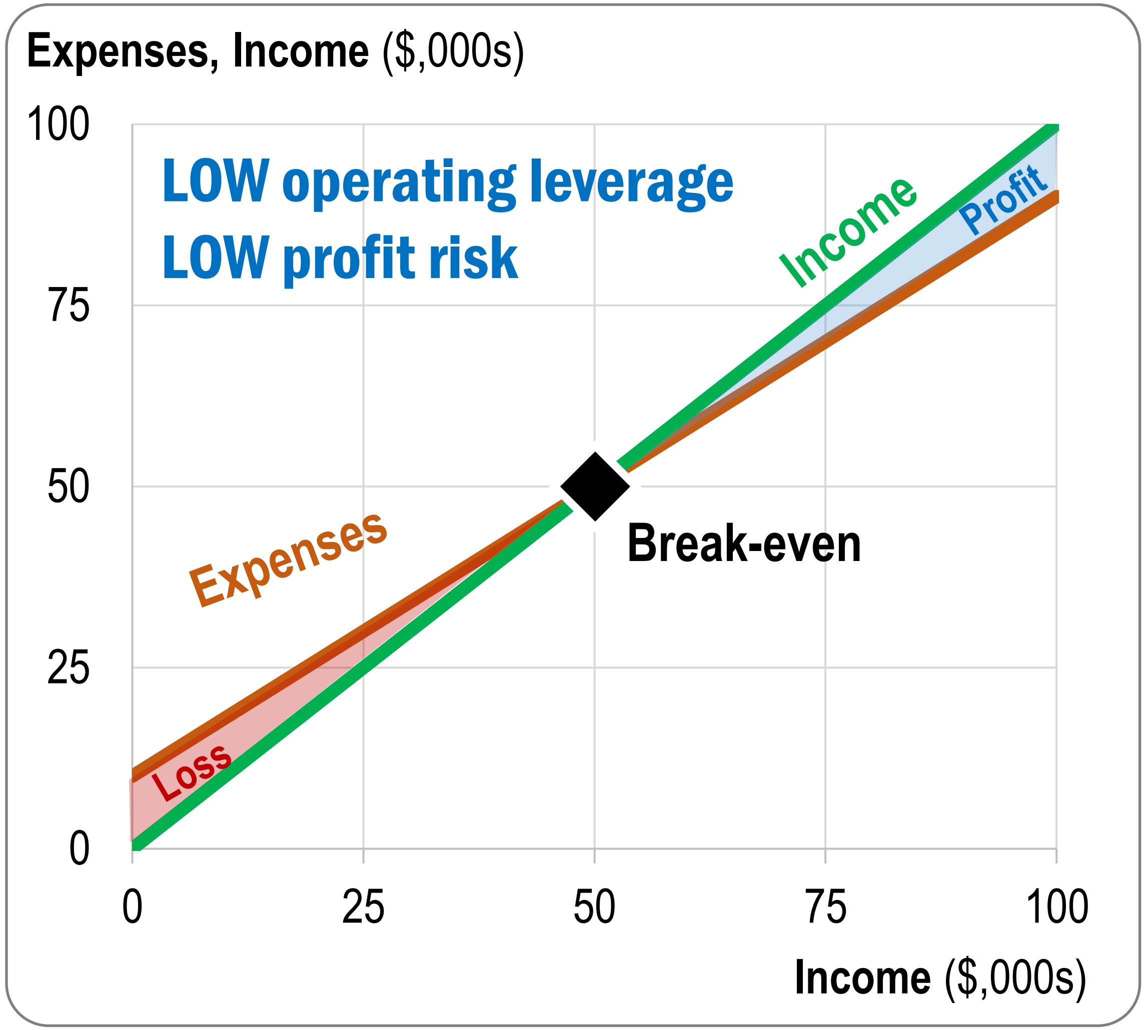
and I shall move the world!"

# Operating Leverage and Risk Visualised

## Fixed Fred




## Flexible Fiona



# Operating Leverage and Risk Calculations

**Fixed Fred**

Expenses = 40,000 + 0.2 × Income

Profit = 0.8 × Income – 40,000

**Flexible Fiona**

Expenses = 10,000 + 0.8 × Income

Profit = 0.2 × Income – 10,000

| | Income: $0 | Income: $50,000 | Income: $100,000 | |
|---|---|---|---|---|
| **Fixed Fred's Profit** | – 40,000 | 0 | + 40,000 | **High Operating Leverage** |
| | | $80,000 variation | | |
| **Flexible Fiona's Profit** | – 10,000 | 0 | + 10,000 | **Low Operating Leverage** |
| | | $20,000 variation | | |

Risk is variation

# Two types of leverage

1. **Operating leverage**

   High fixed expenses in proportion to total expenses increases risk

2. **Financial leverage**

   High fixed <u>interest</u> expenses in proportion to total expenses increases profit risk (income leverage)

   High levels of debt in proportion to total assets increases equity risk (capital leverage)

# Example of leverage using property

**Two investors would like to buy a property for $1,000:**

**Ursa has $1,000 cash and will not borrow anything (unlevered)**

**Liam has $500 cash and will borrow $500 at 5% per annum (levered)**

**Assume no taxes or transaction costs**

**Each investor plans to buy the property then flip (sell) it in 1 year.**

# Example of leverage using property

Two investors would like to buy a property for $1,000:

Ursa has $1,000 cash and will not borrow anything (unlevered)

Liam has $500 cash and will borrow $500 at 5% per annum (levered)

Assume no taxes or transaction costs

Each investor plans to buy the property then flip (sell) it in 1 year.

What is return on equity for the following scenarios:

1. Total return on the asset (property) is 0%?
2. Total return on the asset (property) is 10%?

# Balance sheets

**URSA**

| | | | |
|---|---|---|---|
| **Assets (A)** | **1,000** | **Debt (D)** | **0** |
| | | **Equity (E)** | **1,000** |
| **Value (V)** | **1,000** | **Value (V)** | **1,000** |

$$\mathbf{LVR} = \frac{\mathbf{D}}{\mathbf{V}}$$

**Unlevered**

$$= \frac{\mathbf{0}}{\mathbf{1,000}} = \mathbf{0}$$

LVR is 'Loan to value ration' which is the total loan (debt) divided by the total value of the asset (A) or (V)

# Balance sheets

**URSA**

| | | | |
|---|---|---|---|
| Assets (A) | 1,000 | Debt (D) | 0 |
| | | Equity (E) | 1,000 |
| Value (V) | 1,000 | Value (V) | 1,000 |

$$\mathbf{LVR} = \frac{\mathbf{D}}{\mathbf{V}}$$

Unlevered

$$= \frac{\mathbf{0}}{\mathbf{1,000}} = \mathbf{0}$$

**LIAM**

| | | | |
|---|---|---|---|
| Assets (A) | 1,000 | Debt (D) | 500 |
| | | Equity (E) | 500 |
| Value (V) | 1,000 | Value (V) | 1,000 |

$$\mathbf{LVR} = \frac{\mathbf{D}}{\mathbf{V}}$$

Levered

$$= \frac{\mathbf{500}}{\mathbf{1,000}} = \mathbf{0.5}$$

LVR is ‘Loan to value ration’ which is the total loan (debt) divided by the total value of the asset (A) or (V)

# Ursa's return on equity (unlevered)

| | |
|---|---|
| Return on asset | $r_A = 0\%$ |
| Buy property at t = 0 | –1,000 |
| Sell property at t = 1 | +1,000 |
| Profit | 0 |
| Equity invested | 1,000 |
| Return on equity (profit / equity) | $r_E = 0\%$ |

# Ursa's return on equity (unlevered)

| Return on asset | $r_A = 0\%$ | $r_A = 10\%$ |
|---|---|---|
| Buy property at t = 0 | –1,000 | –1,000 |
| Sell property at t = 1 | +1,000 | +1,100 |
| Profit | 0 | +100 |
| Equity invested | 1,000 | 1,000 |
| Return on equity (profit / equity) | $r_E = 0\%$ | $r_E = +10\%$ |

# Liam's return on equity (levered)

| | |
|---|---|
| Return on asset | $r_A = 0\%$ |
| Buy property at t = 0 | –1,000 |
| Sell property at t = 1 | +1,000 |
| Pay 5% interest on loan | –25 |
| Profit | –25 |
| Equity invested | 500 |
| Return on equity (profit / equity) | $r_E = -5\%$ |

# Liam's return on equity (levered)

| Return on asset | $r_A$ = 0% | $r_A$ = 10% |
|---|---|---|
| Buy property at t = 0 | –1,000 | –1,000 |
| Sell property at t = 1 | +1,000 | +1,100 |
| Pay 5% interest on loan | –25 | –25 |
| Profit | –25 | +75 |
| Equity invested | 500 | 500 |
| Return on equity (profit / equity) | $r_E$ = –5% | $r_E$ = +15% |

# Financial leverage **increases risk**

| | | |
|---|---|---|
| Return on asset | $r_A$ = 0% | $r_A$ = 10% |
| Ursa's return on equity | $r_E$ = 0% | $r_E$ = +10% |
| Liam's return on equity | $r_E$ = –5% | $r_E$ = +15% |

| Return on asset | 0% | 5% | 10% | |
|---|---|---|---|---|
| **Liam return on equity** | – 5% | 5% | + 15% | **High Financial Leverage** |
| | | 20% variation | | |
| **Ursa return on equity** | – 0% | 5% | + 10% | **Low Financial Leverage** |
| | | 10% variation | | |

Risk is variation

# Financial leverage

**Involves borrowing money to buy an investment**

**It magnifies the possible outcomes**

**… which also magnifies risk**

**If the return on the asset exceeds the cost of debt**

**… then financial leverage magnifies profits (return on equity)**

**If the return on asset is less than the cost of debt**

**… then financial leverage magnifies losses (return on equity)**

# Financial leverage is key to wealth creation

**The expected return on a quality property investment**

**… is expected to exceed interest rates over the long-term (20+ years)**

**Financial leverage significantly magnifies**

**… the return on each dollar of your own funds invested (equity)**

**Financial leverage is a key strategy for creating wealth over the long-term**

**… but it does so with a higher level of risk**

# When financial leverage goes wrong …

**Failure to repay debt can cause bankruptcy**

- Close to half of the bankruptcies involves loans less than $15k

**Declaring bankruptcy goes onto the public record**

- … and comes with professional and lifestyle restrictions

**Alternatives to bankruptcy include:**

- Informal arrangement with the lender
- Bankruptcy Act Part IX debt agreement
- Bankruptcy Act Part X arrangement

# Loan Mathematics

# What is an annuity? (recap)

| 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| | **C** | **C** | **C** | **C** | **C** |

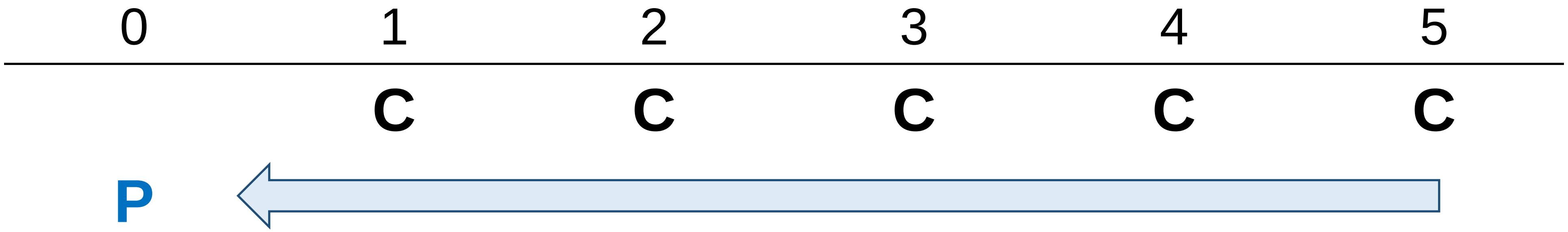
**An annuity is a set of regular cash flows**

**For an ‘ordinary’ annuity**

**… the first cash flow occurs at the end of the first period**

**… and the cash flows are always level (not growing)**

# Present value of an annuity (recap)




**We are calculating the value of the cash flows** (with first cash flow at end of first period)

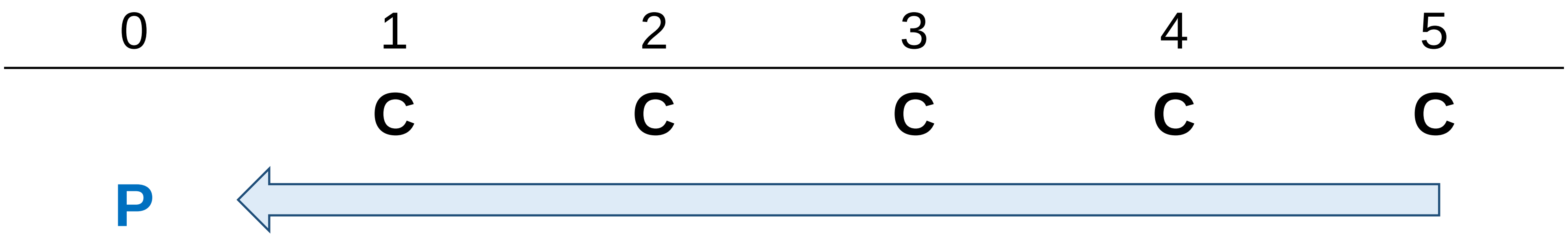
**… at the START of the regular series of payments**

**Here the regular cash flow is 'C'**

**there are 5 cash flows and we are moving them to t = 0**

**to calculate the present value of the annuity 'P'**

# Present value of an annuity formula (recap)




$$P_n = C \times \left[\frac{1-(1+r)^{-n}}{r}\right]$$

**'C' are the regular cash flows that are being moved to the left**

**'n' is the number of cash flows** (with first cash flow at end of first period)

**'r' is the effective rate of return per period** (time value of money)

**'P' is value of the regular series of 'n' cash flows of 'C' at the start (left)**

# Borrowing capacity example

| 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| | **10** | **10** | **10** | **10** | **10** |

**P**

$$\mathbf{P_n = C \times \left[\frac{1-(1+r)^{-n}}{r}\right]}$$

**How much we borrow if we can pay $10 p.a. for 5 years at 5% p.a.?**

*** By default you should assume that the cash flows occur at the end of each period (unless otherwise specified)**

# Borrowing capacity example

| 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| | **10** | **10** | **10** | **10** | **10** |

**P**

$$\mathbf{P_n = C \times \left[\frac{1-(1+r)^{-n}}{r}\right]}$$

**How much we borrow if we can pay $10 p.a. for 5 years at 5% p.a.?**

$$\mathbf{P_n = 10 \times \left[\frac{1-1.05^{-5}}{0.05}\right]}$$

$$\mathbf{= \$43.29}$$

**Calculator**

$$\mathbf{10 \times (1 \; - \; 1.05 \; x^{\blacksquare} \; - 5 \rightarrow) \div 0.05 \; =}$$

**Spreadsheet**

$$\mathbf{= 10 * ( \; 1 \; - \; 1.05 \; \wedge - 5 \,)/ \; 0.05}$$

*** By default you should assume that the cash flows occur at the end of each period (unless otherwise specified)**

# Loan repayment example

| 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| | C | C | C | C | C |

**$43.29** ←

$$\mathbf{C = P_n \div \left[\frac{1-(1+r)^{-n}}{r}\right]}$$

**We borrow $43.29 now. Yearly payments for 5 years at 5% p.a.?**

*** By default you should assume that the cash flows occur at the end of each period (unless otherwise specified)**

# Loan repayment example

| 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| | C | C | C | C | C |

**$43.29** ←

$$\mathbf{C = P_n \div \left[\frac{1-(1+r)^{-n}}{r}\right]}$$

**We borrow $43.29 now. Yearly payments for 5 years at 5% p.a.?**

$$\mathbf{C = 43.29 \div \left[\frac{1-1.05^{-5}}{0.05}\right]}$$

$$\mathbf{= \$10.00}$$

**Calculator**

$$\mathbf{43.29 \div ( ( 1 - 1.05\ x^{\blacksquare} - 5 \rightarrow ) \div 0.05 ) =}$$

**Spreadsheet**

$$\mathbf{= 43.29 / ( ( 1 - 1.05\ \wedge - 5 ) / 0.05 )}$$

*** By default you should assume that the cash flows occur at the end of each period (unless otherwise specified)**

# How much can David borrow?

**David is able to afford monthly loan repayments of $1,000 per month. He is considering taking a home loan with a loan term of 25 years (300 months) at an interest rate of 6% per year compounded monthly (0.5% per month). The first loan repayment would be in one month. How much can he borrow?**

$$\mathbf{P_n = C \times \left[\frac{1-(1+r)^{-n}}{r}\right]}$$

# How much can David borrow?

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$$\mathbf{P_n = C \times \left[\frac{1-(1+r)^{-n}}{r}\right]}$$

$$\mathbf{= 1,000 \times \left[\frac{1-1.005^{-300}}{0.005}\right]}$$

$$\mathbf{= \$155,207}$$

# What will be Fiona's loan repayments?

Fiona would like to borrow $400,000 to buy a new home. She is considering a loan term of 30 years (360 months) at an interest rate of 6% per year compounded monthly (0.5% per month). The first loan repayment would be in 1 month. What will be her monthly loan repayments?

$$\mathbf{C = P_n \div \left[\frac{1-(1+r)^{-n}}{r}\right]}$$

# What will be Fiona's loan repayments?

**Fiona would like to borrow $400,000 to buy a new home. She is considering a loan term of 30 years (360 months) at an interest rate of 6% per year compounded monthly (0.5% per month). The first loan repayment would be in 1 month. What will be her monthly loan repayments?**

$$\mathbf{C = P_n \div \left[\frac{1-(1+r)^{-n}}{r}\right]}$$

$$\mathbf{= 400{,}000 \div \left[\frac{1-1.005^{-360}}{0.005}\right]}$$

$$\mathbf{= \$2{,}398}$$

# Interest Rates

# Interest Rates driven by Monetary Policy

**Reserve Bank of Australia (RBA) influences interest rates**

**Monetary policy involves:**

1. **Setting the ‘cash rate’ which directly influences all interest rates**
2. **Buying and selling government bonds to influence liquidity**

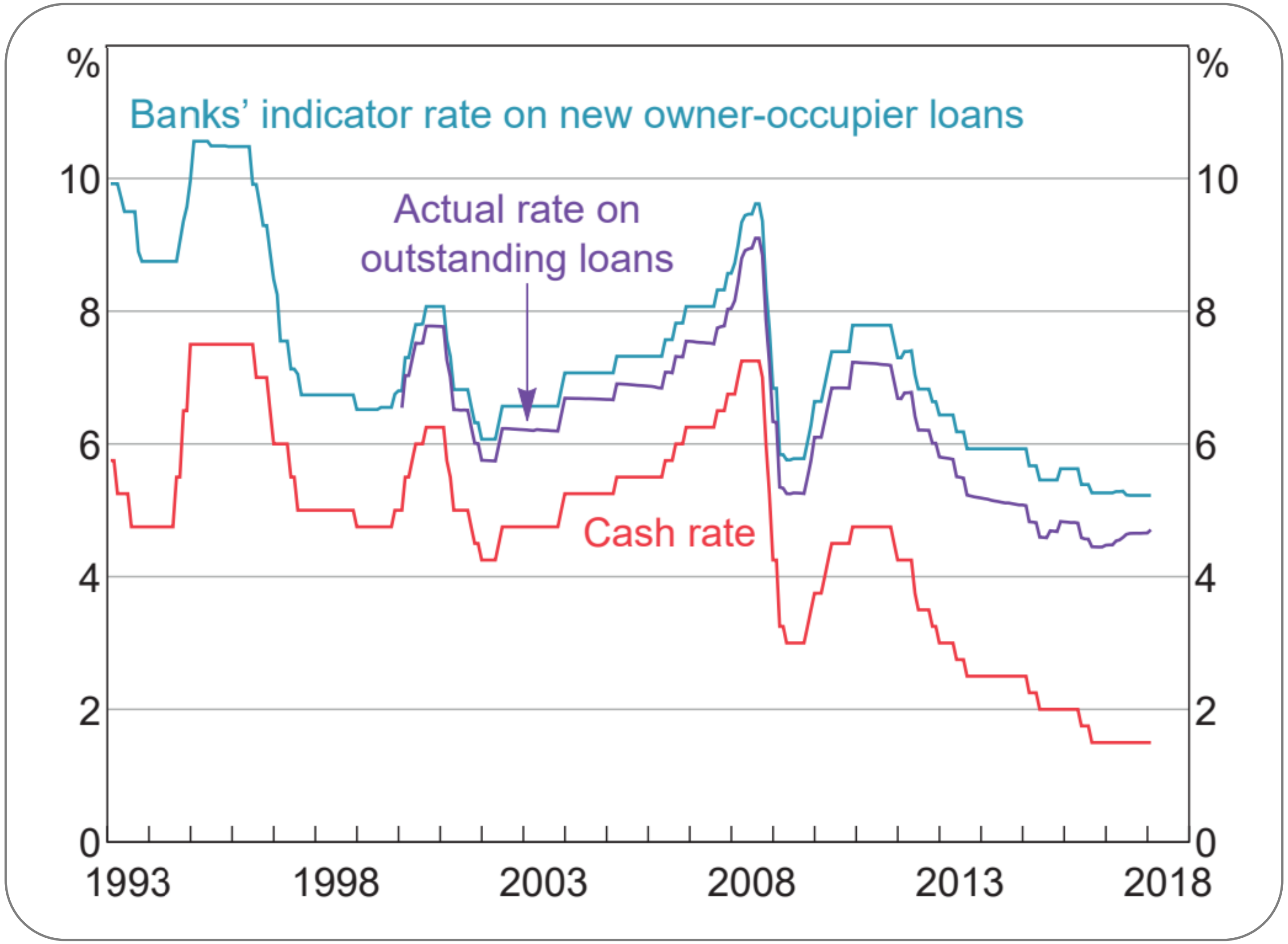
**The goal of this is to influence aggregate expenditure in the economy**

- Higher interest rates decrease consumption and investment
- Lower interest rates can stimulate consumption and investment

**The level of aggregate expenditure influences inflation**

**RBA target for inflation is between 2% – 3% p.a.**

# Cash rate affects all other interest rates




Source: Reserve Bank of Australia (RBA)

# Interest rate components

| | |
|---|---|
| Risk | Risk of the borrowers failing to make repayments |
| Market Power | Power of financial institutions over borrowers and lenders (competitive rivalry) |
| Loanable Funds | Demand and supply of loanable funds |
| Real Cash Rate | Central bank normally sets nominal cash rate above expected inflation |
| Inflation | Reward for lending should exceed inflation |

# How do banks determine fixed interest rates?

**Fixed interest rate = average expected variable rate + liquidity premium**

**Expected variable rates are 3% in 1st year, 4% in 2nd year and 5% in 3rd year and 0.5% liquidity premium. What is 3 year fixed interest rate?**

| 0 | | 1 | | 2 | | 3 |
|---|---|---|---|---|---|---|
| | 3% | | 4% | | 5% | |

**Calculate average***

$$\bar{i} = \frac{i_1 + i_2 + i_3}{3}$$

$$= \frac{3\% + 4\% + 5\%}{3} = 4\%$$

**Then add liquidity premium**

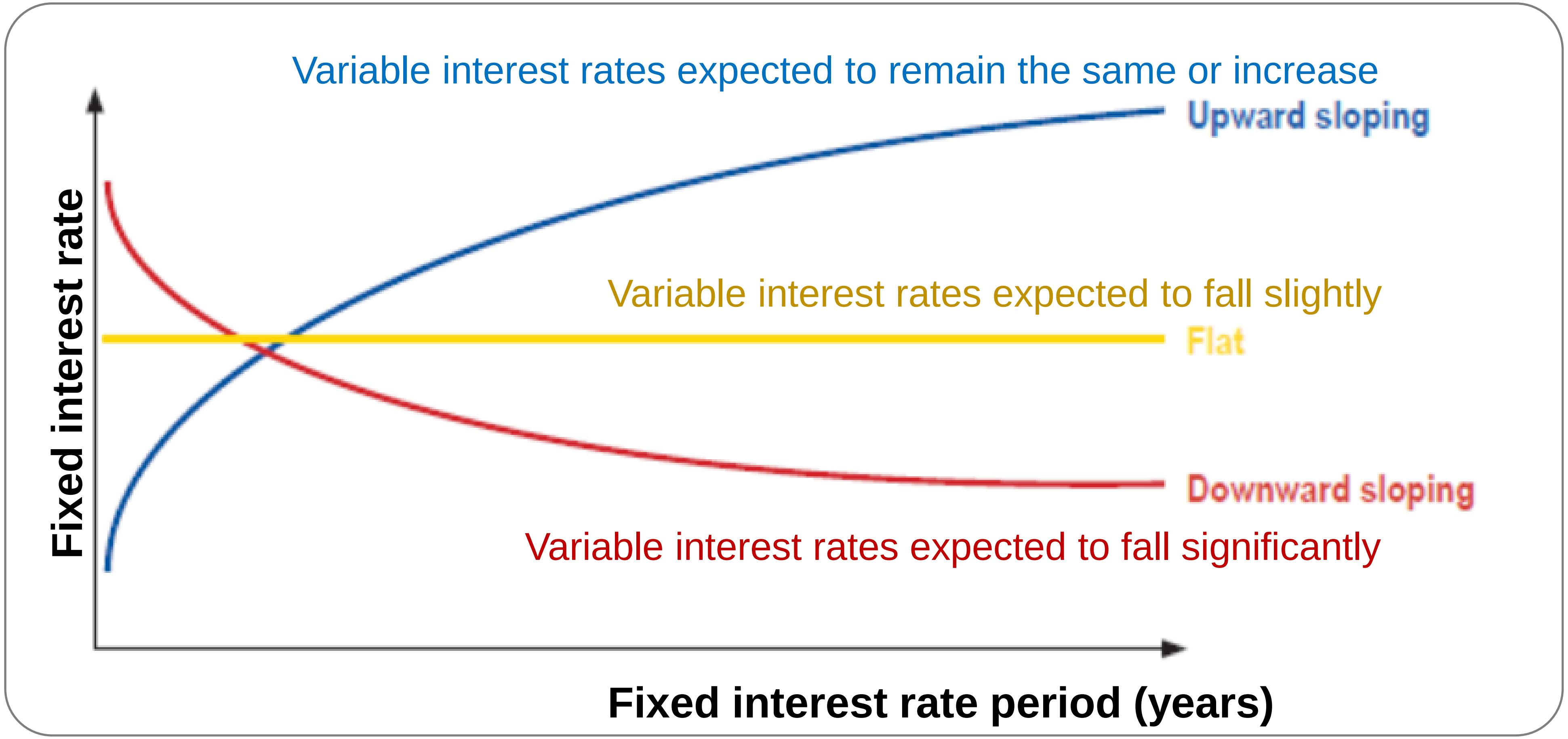
$$i_f = 4\% + 0.5\%$$

$$= 4.5\%$$

* A geometric average is normally used but I've used an arithmetic average here for simplicity!

# Term structure of interest rates and yield curves

## Fixed interest rate = average expected variable rate + liquidity premium




In bond markets, this is called a 'yield curve' and shows the relationship between the maturity of a bond and the yield to maturity of the bond
Fixed interest rates for different fixed periods are determined by bond market yields.

# Interest rate matrix

**Interest only attracts a premium over principal and interest loans**

**Investment loans attract a premium over home loans**

| | Principal + Interest | Interest Only |
|---|---|---|
| **Home Loan** | **6.74%** | **7.29%** +0.55% |
| **Investment Loan** | **7.34%** +0.60% | **7.59%** +0.85% |

# Comparison rates

**Interest rate = Interest cost per year**

**Comparison rate = Interest + up-front and ongoing fees per year**

**Calculated based on $150,000 loan over a 25-year loan term**

| | Interest Rate | Comparison Rate |
|---|---|---|
| Home Loan | 6.74% | 6.84% +0.10% |

# Loan Features

# 3 types of lenders

1. **Major retails banks**

   ANZ (ING Direct), Westpac, St George, CBA, NAB (UBank)

   Good service and good features

   Consistent, moderate interest rates

2. **Smaller banks and lenders**

   Adelaide Bank, Bendigo Bank, Bank of Queensland

   Can offer low interest rates to make sales … but then increase the rate once you are in

3. **Mortgage brokers and other non-bank financial inst.**

   Aussie Home Loans, Mortgage Choice

   Help you to find the cheapest home loan

# 2 repayment options

1. **Payments consist of both principal and interest**

   Each loan payment is part principal and interest (mostly interest towards start of loan)

   Loan is fully paid off at the end of the term (eg 25 years)

2. **Interest only payments**

   Only pay interest on principal for first X years (10 years max)

   Reverts to principal and interest loan after X years

   Popular with owners of investment properties who want to maximise tax deductible interest

# 3 interest rate options

1. **Variable interest rates**

   Monthly repayments change as variable interest rates change

   Expect average variable rate is less than fixed rate over long-run (liquidity premium)

2. **Fixed interest rates**

   Calculated as average of expected future variable rates plus liquidity premium

   Maximum fixed rate period offered by banks is usually 10 years

   Can be charged a large penalty fee if re-finance loan within fixed period

3. **Split rates**

   Part variable and part fixed … people use them to ‘hedge their bets’

   Attractive if you want a fixed rate but also some the flexibility of variable products (redraw)

# Repayment terms

**Period over which loan must be repaid**

**Maximum is usually 30 years**

**Shorter the term**

- Higher the compulsory repayments
- The lower the total interest in the long-run
- Less likely to be able to make voluntary extra repayments
- Choose if you need the bank to force you to make high payments

**Longer the term**

- Lower the compulsory repayments but higher interest
- Choose if you are a good saver and will make extra repayments

# Mortgage offsets and redraw facilities

## Mortgage offset account

- Balance of your savings account reduces principal outstanding
- If average balance is $10,000 then …
- … bank calculates interest on loan outstanding - $10,000
- Allows you to pay off loan faster with same repayments
- Very good feature if it doesn't cost you a higher interest rate!

## Redraw facility

- Allows you to withdraw extra repayments at any time

## Normally only available with variable rate loans

- Some fixed interest loans offer limited redraw or offset facilities

# Family Security Guarantee

**A family member adds their own property as security on your loan**

**Total assets as collateral often doubles**

**Bank may be willing to lend 100% of value of the property**

**No Lenders Mortgage Insurance (LMI) since LVR is under 80%**

**Your parent’s property is at risk if you default**

**… but this is unlikely if you saved 20% deposit**

**… and place the entire deposit on ‘redraw’ on day 1**

# Reverse Mortgage or Home Equity Access

**Reverse Mortgage allows aged 60+ to borrow against their home equity**

… and make no loan repayments for the rest of their life

**When property is eventually sold (possibly upon death)**

… bank claims the ‘principal and interest’ out of the proceeds of the sale

**There may be little left for children or grandchildren**

… which may be useful if you have no children (or don’t like them!)

**Federal Government’s Home Equity Access Scheme**

A more compassionate system that gives you access to equity in home

… on much better terms than a Reverse Mortgage from a financial institution

Services Australia (2024), ‘Home Equity Access Scheme’ at servicesaustralia.gov.au

# Home Loan Tips

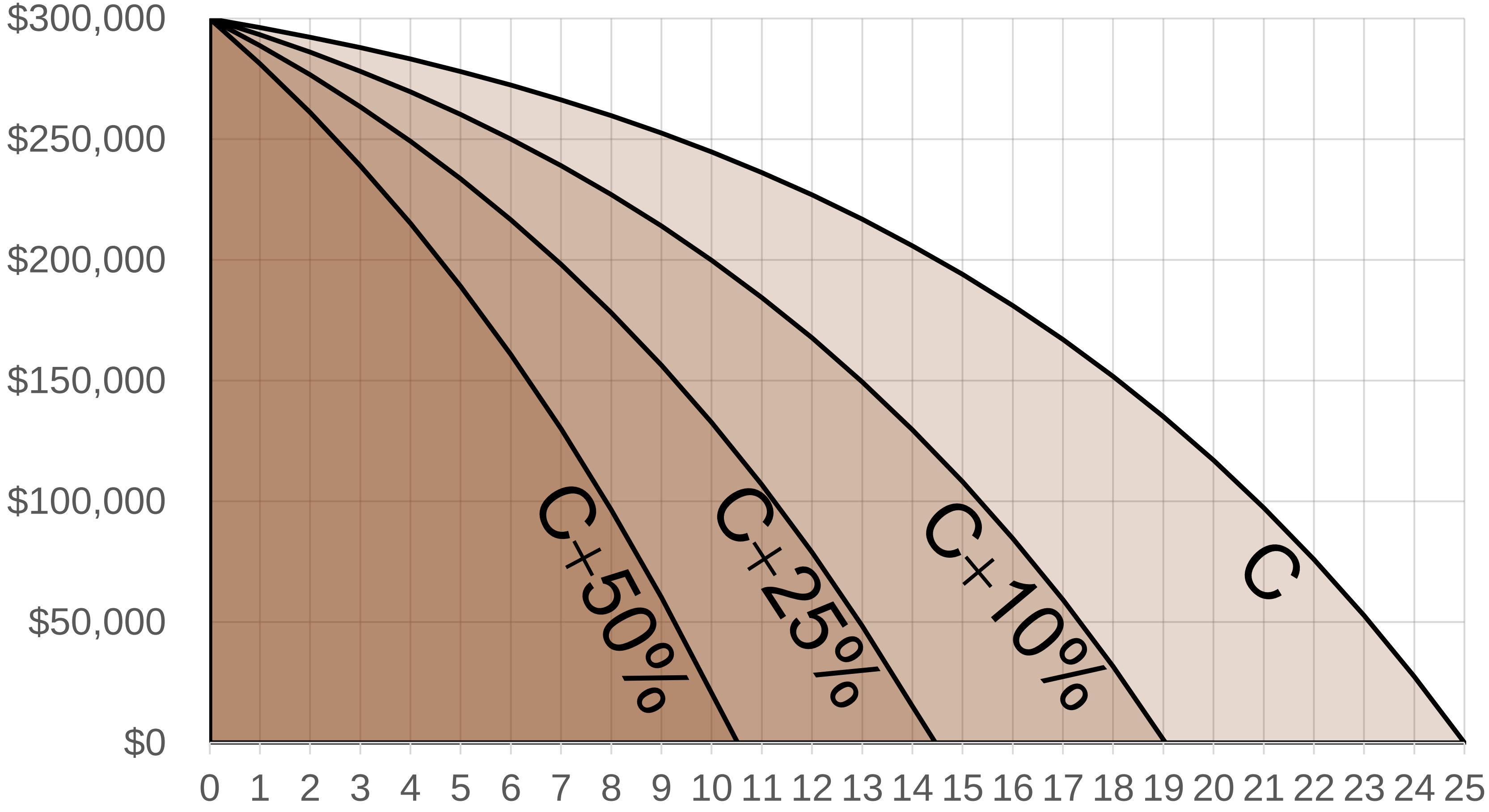
# 1. Pay more than the minimum required

This is usually only an option if you have a variable or split loan

Any additional payment are available in redraw and can still be accessed

# Accelerated loan repayments

**The way out of slavery!**



# 2. Avoid borrowing ‘to capacity’

# 3. Avoid paying Lenders Mortgage Insurance

**Borrow less than 80% or use a Family Security Guarantee**

# 4. Use Family Security Guarantee if possible

**Borrow 100% of value of property**

**Place your 20% deposit on 'redraw'**

**You still have access to these funds**

**… for emergency or to make other investments**

# 5. Increase your payment frequency to weekly

**This will make a slight difference if you don’t have a mortgage offset**

# 6. Move all your savings into redraw

Any savings in the redraw facility can be accessed within 24 hours

They effectively earn the home loan’s rate of interest (tax free)

# 7. Be careful of too much in offset account

**Keeping $100,000 in an offset account is the same as $100,000 in redraw**

**… but you will feel ‘richer’ every time you see your transaction balance**

**… and will end up spending a lot more!**

# 8. Negotiate a good interest rate

**Shop around, use a mortgage broker**

**… threaten to refinance with another bank!**

# 9. Variable rates should be better over long-run

You are saving the ‘liquidity premium’

… and have a lot more flexibility to make additional repayments

… and refinance the home loan (lower rates)

# 10. Positive cash flow v negative gearing

**Australians seem to be obsessed with paying less tax (negative gearing)**

**Negatively geared properties are usually negative cash flow**

… although not always due to deprecation tax deductions <u>not</u> being a cash flow

**Positive cash flow investments = less cash flow stress and happier life!**

# Your Financial Plan

**What is your long-term strategy for property?**

**Will you focus on only your primary residence or investment properties too?**

**Perform some detailed research for your next property acquisition**

**What will be your ‘dwelling’ needs for each of your life stages**

**What will be your borrowing strategy for each life-stage?**

**What product features are attractive to you? Why?**

**More information under ‘Financial Plan Instructions’ document**

… on the course website under ‘Financial Plan’ section

# Join us in a Drop-in Session

**Each week we have a drop-in Q&A session on Zoom**

**Details are on the course website under ‘Getting Started Hub’**

**Bring along some questions that you have about the Unit.**

**Sessions are <u>not</u> recorded.**

**You are also welcome to ask questions on the ‘General Forums’**